

11001 微積分乙期中考評分標準

題號 1. (a)

題目: $f(x) = x^4 + 4x^2 + 5$, $x > 0$, 證明 $f(x)$ 是否為一對一函數?

解法一:

4分 $\{ b \neq a, f'(x) = 4x^3 + 8x > 0, \text{ 當 } x > 0.$

由均值定理可知,

$f(b) - f(a) = f'(c)(b-a)$, 因為 $f'(x) > 0$, 所以會有以下兩種情況 =

1分

$$\begin{cases} b > a & f(b) > f(a) \\ b < a & f(b) < f(a) \end{cases}$$

由這兩種情況可知, $b \neq a \Rightarrow f(b) \neq f(a)$.

$\therefore f$ 為 1-1.

解法二:

1分 $\{ f(a) = f(b), \text{ 假設 } a \neq b,$

依 Rolle's theorem, $\exists c \in (a, b), f'(c) = 0.$

因為 $f'(x) > 0$, 所以 $a \neq b$ (~~不~~)

1分 $\Rightarrow a = b$

$\therefore f$ 為 1-1

解法三:

$$f(x) = (x^2 + 2)^2 + 1$$

$$f(x_1) = f(x_2) \Rightarrow (x_1^2 + 2)^2 + 1 = (x_2^2 + 2)^2 + 1$$

1分

$$\Rightarrow (x_1^2 + 2)^2 = (x_2^2 + 2)^2$$

$$\Rightarrow x_1^2 + 2 = x_2^2 + 2$$

$$\Rightarrow x_1^2 = x_2^2$$

$$\Rightarrow \underline{x_1 = x_2} \quad (\because x_1 > 0, x_2 > 0)$$

$\therefore f$ 為 1-1. 1分

3分

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題號

16.

法 1.

$$x^4 + 4x^2 + 5 = y \Rightarrow (x^2 + 2)^2 = y - 1 \Rightarrow x^2 + 2 = \pm\sqrt{y-1} \quad \left(\begin{array}{l} \because x^2 + 2 > 0 \\ \therefore \text{負不合} \end{array} \right)$$

$$\Rightarrow x^2 = -2 + \sqrt{y-1} \Rightarrow x = \pm\sqrt{-2 + \sqrt{y-1}} \quad \left(\begin{array}{l} \because x > 0 \\ \therefore \text{負不合} \end{array} \right)$$

$$\Rightarrow x = \sqrt{-2 + \sqrt{y-1}} \Rightarrow f^{-1}(x) = \sqrt{-2 + \sqrt{x-1}}$$

條件 1

法 2. 令 $g(x) = f^{-1}(x)$

$$f(g(y)) = y \Rightarrow [g(y)]^4 + 4g(y) + 5 = y \Rightarrow [g(y)]^2 + 2 = \pm\sqrt{y-1}$$

$$\Rightarrow [g(y)]^2 + 2 = \pm\sqrt{y-1} \quad \left(\begin{array}{l} \because [g(y)]^2 + 2 > 0 \\ \therefore \text{負不合} \end{array} \right) \Rightarrow [g(y)]^2 = -2 + \sqrt{y-1}$$

$$\Rightarrow g(y) = \pm\sqrt{-2 + \sqrt{y-1}} \quad \left(\begin{array}{l} \because g(y) > 0 \\ \therefore \text{負不合} \end{array} \right) \Rightarrow g(y) = \sqrt{-2 + \sqrt{y-1}}$$

$$\Rightarrow f^{-1}(x) = \sqrt{-2 + \sqrt{x-1}}$$

條件 1

法 3.

$$x^4 + 4x^2 + 5 = y \Rightarrow (x^2)^2 + 4(x^2) + 5 - y = 0$$

$$\Rightarrow x^2 = \frac{-4 \pm \sqrt{4^2 - 4(5-y)}}{2} = \frac{-4 \pm \sqrt{4y-4}}{2} = -2 \pm \sqrt{y-1} \quad \left(\begin{array}{l} \because x^2 > 0 \\ \therefore \text{負不合} \end{array} \right)$$

$$\Rightarrow x^2 = -2 + \sqrt{y-1} \Rightarrow x = \pm\sqrt{-2 + \sqrt{y-1}} \quad \left(\begin{array}{l} \because x > 0 \\ \therefore \text{負不合} \end{array} \right)$$

$$\Rightarrow x = \sqrt{-2 + \sqrt{y-1}} \xrightarrow{x, y \text{ 互換}} y = \sqrt{-2 + \sqrt{x-1}}$$

條件 1

扣分標準 1. 答案錯則扣 2 分.

2. 沒有寫出 條件 1 則扣 1 分.

3. 無計算過程, 但答案正確扣 2 分.

4. $x > 0$ 的條件, 題目有給, 沒寫不扣分.

若整題只有將 x, y 互換
寫出 $x = y^4 + 4y^2 + 5$
則只算 1 分.

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題號 C

法 1.

由課本可知 $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$ 故 $(f^{-1})'(10) = \frac{1}{f'(f^{-1}(10))}$ 只有寫出此式得 1 分

求 $f^{-1}(10) = c$

法 1.1 由 1b $f^{-1}(10) = \sqrt{-2 + \sqrt{10-1}} = \sqrt{-2+3} = 1$

法 1.2 $f(c) = 10 \Rightarrow x^4 + 4x^2 + 5 = 10$

$$\Rightarrow (x^2)^2 + 4(x^2) - 5 = 0$$

$$\Rightarrow (x^2 + 5)(x^2 - 1) = 0$$

$$\Rightarrow x = 1 \text{ or } -1 (\because x > 0, \therefore -1 \text{ 不合})$$

法 1.3 $\because f(1) = 10 \therefore f^{-1}(10) = 1$

$$f'(x) = 4x^3 + 8x$$

$$\Rightarrow (f^{-1})'(10) = \frac{1}{f'(f^{-1}(10))} = \frac{1}{f'(1)} = \frac{1}{4 \times 1^3 + 8 \times 1} = \frac{1}{12}$$

答案錯扣 2 分

法 2.

$$f^{-1}(x) = \sqrt{-2 + \sqrt{x-1}} = ((x-1)^{\frac{1}{2}} - 2)^{\frac{1}{2}}$$

$$(f^{-1})'(x) = \frac{1}{2} ((x-1)^{\frac{1}{2}} - 2)^{-\frac{1}{2}} \cdot \frac{1}{2} \cdot (x-1)^{-\frac{1}{2}} \cdot 1$$

上半部

微分正確則 2 分

$$(f^{-1})'(10) = \frac{1}{2} ((10-1)^{\frac{1}{2}} - 2)^{-\frac{1}{2}} \cdot \frac{1}{2} \cdot (10-1)^{-\frac{1}{2}} \cdot 1$$

$$= \frac{1}{2} (3-2)^{-\frac{1}{2}} \cdot \frac{1}{2} \cdot 9^{-\frac{1}{2}} \cdot 1$$

$$= \frac{1}{2} \cdot \frac{1}{1} \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot 1 = \frac{1}{12}$$

下半部佔 3 分

答案有錯扣 2 分

計算佔 1 分

2(a) 共5分

$$\lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3}{x-2} = \lim_{x \rightarrow 2} \frac{(\sqrt{x+7} - 3)(\sqrt{x+7} + 3)}{(x-2)(\sqrt{x+7} + 3)} \rightarrow \text{有理化: 2分}$$

$$= \lim_{x \rightarrow 2} \frac{x+7-9}{(x-2)(\sqrt{x+7} + 3)} \left. \begin{array}{l} \text{化簡: 1分} \end{array} \right\}$$

$$= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x+7} + 3}$$

$$= \frac{1}{6} \rightarrow \text{答案: 2分}$$

* 若化簡、有理化過程錯誤
答案正確亦不給分

2a

共5分

$$\lim_{x \rightarrow 0} \frac{\tan 5x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{\sin 5x}{\cos 5x} \cdot \frac{1}{\sin 3x} \rightarrow 1分$$

$$= \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \frac{5x}{\cos 5x} \cdot \frac{3x}{\sin 3x} \cdot \frac{1}{3x} \left. \begin{array}{l} \text{變形: 2分} \end{array} \right\}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \frac{5x}{3x} \cdot \frac{1}{\cos 5x} \cdot \frac{3x}{\sin 3x}$$

$$= 1 \times \frac{5}{3} \times 1 \times 1 = \frac{5}{3} \rightarrow \text{答案: 2分}$$

* 若使用 $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$ 而未詳加說明, 扣1分 (by 出題老師)

* 若過程有明顯錯誤而獲致正確答案亦不給分

* 若答案正確而過程說明不夠清楚, 酌扣過程分數 (但可得到答案分數)

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題號 2c/2d

$$(2c) \lim_{x \rightarrow -\infty} x + \sqrt{x^2+3} = \lim_{x \rightarrow -\infty} \frac{(x + \sqrt{x^2+3})(x - \sqrt{x^2+3})}{x - \sqrt{x^2+3}} \quad (2 \text{ points})$$

$$= \lim_{x \rightarrow -\infty} \frac{x^2 - (x^2+3)}{x - \sqrt{x^2+3}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3}{x - \sqrt{x^2+3}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3}{x - |x| \sqrt{1 + \frac{3}{x^2}}} \quad (2 \text{ points})$$

$$= \lim_{x \rightarrow -\infty} \frac{-3}{x - (-x) \sqrt{1 + \frac{3}{x^2}}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3}{x + x \sqrt{1 + \frac{3}{x^2}}}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3}{x(1 + \sqrt{1 + \frac{3}{x^2}})}$$

$$= 0 \quad (1 \text{ point})$$

$$(2d) \lim_{x \rightarrow \infty} \frac{\sqrt[3]{8x^6+2x^3+1}}{\sqrt{4x^4+3x+1}}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt[3]{8x^6(1 + \frac{2x^3}{8x^6} + \frac{1}{8x^6})}}{\sqrt{4x^4(1 + \frac{3x}{4x^4} + \frac{1}{4x^4})}} \quad (2 \text{ points})$$

$$= \lim_{x \rightarrow \infty} \frac{2x^2 \sqrt[3]{1 + \frac{1}{4x^3} + \frac{1}{8x^6}}}{2x^2 \sqrt{1 + \frac{3}{4x^3} + \frac{1}{4x^4}}}$$

$$= 1 \quad (1 \text{ point})$$

(2 points for implying $\frac{1}{\infty} = 0$)

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題號

3

$$(a) \quad f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

2分

(寫切線斜率不給分、lim 沒寫不給分)

$$(b) \quad \lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$$

4分

$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

$$f(0) = |0| = 0$$

$$\therefore \lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^-} |x| = 0 = f(0) \quad (\text{沒寫扣1分})$$

$\therefore f$ is conti at $x_0 = 0$

$$4分 \quad \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{|x| - 0}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{|x| - 0}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$$

$$\therefore \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \text{ doesn't exist}$$

$\therefore f$ is not diff. at $x_0 = 0$

(畫圖說故事不給分、寫尖黑點不可微不給分)

(只寫 $\lim_{x \rightarrow 0^+} f'(x) \neq \lim_{x \rightarrow 0^-} f'(x)$ 不給分, 要算出 1, -1)

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題號 4a

$$y = (\sin x)^{\cos x}$$

解法一：

$$\ln y = \cos x \ln \sin x$$

—— 2pts

$$\Rightarrow y'/y = -\sin x \ln \sin x + \cos^2 x / \sin x$$

$$\Rightarrow y' = y \left(\underbrace{-\sin x \ln \sin x}_{1pt} + \underbrace{\cos^2 x / \sin x}_{1pt} \right)$$

$$= (\sin x)^{\cos x} (-\sin x \ln \sin x + \cos^2 x / \sin x) \quad - 1pt$$

解法二：

$$y = (\sin x)^{\cos x} = e^{\cos x \ln \sin x}$$

—— 2pts

$$\Rightarrow y' = (e^{\cos x \ln \sin x}) \left(\underbrace{-\sin x \ln \sin x}_{1pt} + \underbrace{\cos^2 x / \sin x}_{1pt} \right)$$

$$= (\sin x)^{\cos x} (-\sin x \ln \sin x + \cos^2 x / \sin x) \quad - 1pt$$

其餘錯誤的地方扣1分

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題號 4(b)(c)

(b).

$$y = \ln\left(\frac{e^x-1}{e^x+1}\right)$$

法 1:

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\left(\frac{e^x-1}{e^x+1}\right)} \cdot \frac{e^x(e^x+1) - e^x(e^x-1)}{(e^x+1)^2} \\ &= \frac{e^x+1}{e^x-1} \cdot \frac{e^x+e^x - e^{2x}+e^x}{(e^x+1)^2} \\ &= \frac{2e^x}{e^{2x}-1} \quad \# \end{aligned}$$

寫出 $\frac{1}{\frac{e^x-1}{e^x+1}}$ or $\frac{e^x+1}{e^x-1}$... 2分.

$\frac{e^x(e^x+1) - e^x(e^x-1)}{(e^x+1)^2}$ or $\frac{2e^x}{(e^x+1)^2}$... 2分.

$\frac{2e^x}{e^{2x}-1}$... 1分.

法 2:

$$y = \ln\left(\frac{e^x-1}{e^x+1}\right) = \ln(e^x-1) - \ln(e^x+1)$$

寫出 $y = \ln(e^x-1) - \ln(e^x+1)$... 1分.

$$\begin{aligned} \frac{dy}{dx} &= \frac{e^x}{e^x-1} - \frac{e^x}{e^x+1} \quad (\text{未整理也給對}) \\ &= \frac{2e^x}{e^{2x}-1} \quad \# \end{aligned}$$

$\frac{e^x}{e^x-1}$... 2分.

$\frac{e^x}{e^x+1}$... 2分.

(c).

$$y = \arctan \frac{x}{2} - \frac{1}{2(x^2+4)}$$

寫出 $\frac{1}{1+(\frac{x}{2})^2}$... 2分.

$$\begin{aligned} \frac{dy}{dx} &= \frac{\frac{1}{2}}{1+(\frac{x}{2})^2} - \frac{-2 \cdot 2x}{[2(x^2+4)]^2} \\ &= \frac{2}{4+x^2} + \frac{x}{(x^2+4)^2} \end{aligned}$$

$\frac{-2 \cdot 2x}{[2(x^2+4)]^2}$ or $\frac{-x}{(x^2+4)^2}$... 2分.

$\frac{2x^2+x+8}{(x^2+4)^2}$... 1分.

$$= \frac{2(x^2+4) + x}{(x^2+4)^2}$$

$$= \frac{2x^2+x+8}{(x^2+4)^2} \quad \#$$

註: 1. 答案未整理, 一樣給分.

2. 過程皆對, 答案錯誤皆扣1分.

3. 分段給分, 若正負號錯誤則不給分.
計及錯誤.

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題號 5

$$x^3 + ye^x + y^3 = 2 \quad \text{對} x \text{微分}$$

$$3x^2 + \frac{dy}{dx}(e^x + 3y^2) + ye^x = 0$$

$$\frac{dy}{dx} = -\frac{3x^2 + ye^x}{e^x + 3y^2} \quad \text{代入}(0,1)$$

$$\frac{dy}{dx} = -\frac{1}{1+3} = -\frac{1}{4}$$

$$\text{切線方程式: } y = -\frac{1}{4}x + 1$$

- (1) 隱微分共 3 分，部分微分錯誤扣 2 分，正負號錯誤扣 1 分。
- (2) 斜率求出共 1 分。
- (3) 切線方程式共 2 分，算錯但有試著求出切線方程式給 1 分，(1)、(2)皆錯誤的情況下不給分。

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題號

6

① $f(x) = x^{\frac{1}{4}}$ (1分) , ② $x_0 = 625$ (1分) , ③ $\Delta x = 1$ (1分)

④ $f'(x) = \frac{1}{4} x^{-\frac{3}{4}}$ (1分)

$$f'(625) = \frac{1}{4} \cdot 625^{-\frac{3}{4}} = \frac{1}{4} \cdot \frac{1}{5^3} = \frac{1}{500} = 0.002$$

$$\begin{aligned} f(x+\Delta x) &\approx f(x) + f'(x) \Delta x \\ &= f(625) + f'(625) \cdot 1 \\ &= 5 + 0.002 \cdot 1 \end{aligned}$$

$$\therefore \sqrt[4]{626} = 5.002 \quad (2分)$$

(算對答案即可給6分)

(算錯答案按照以上配分方式給分)

($\sqrt[4]{625}$ 寫成 25 只扣 1 分、 $\frac{1}{500}$ 寫成 0.2 只扣 1 分)

(答案寫 $5 + \frac{1}{500}$ 、 $\frac{2501}{500}$ 都給分)

(①②③④ 寫對, 但 $f'(625)$ 代成 $f'(1)$ 不扣分)

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題號

2

$$\text{Let } f(x) = 2x - 2 - \cos x$$

存在性: $f(0) = -3, f(\pi) = 2\pi - 1 > 0$

(3分)

$\therefore -3 < 0 < 2\pi - 1$ & f is continuous everywhere.

$\therefore$ By IVT, $\exists c \in (0, \pi)$ s.t. $f(c) = 0$ — ①

唯一性: Suppose that $f(x)$ has two distinct roots a, b ($a < b$)
(3分) such that $f(a) = f(b) = 0$

$a, b \in \mathbb{R}$

範圍錯扣1分

$\therefore f$ is continuous and differentiable everywhere.

$\therefore$ By Rolle's Theorem, $\exists p$ in (a, b) such that $f'(p) = 0$

But $f'(x) = 2 + \sin x > 0 \Rightarrow f'(p) \neq 0$ ($\rightarrow \leftarrow$)

$\therefore f(x) = 2x - 2 - \cos x = 0$ has exactly one solution.

也可用嚴格遞增: $f'(x) = 2 + \sin x > 0$

$\Rightarrow f$ is strictly increasing. — ②

由①②兩條件可知 $f(x) = 0$ 只有一解

• $f'(x)$ 算錯扣1分.

• 條件有寫, 沒寫定理名稱不扣分.

兩者都沒寫扣1分.

• 說明不完整扣1分.

• 只說明嚴格遞增

未說明有解扣3分.

• 區間範圍寫錯扣1分.

• f 的 domain 寫錯扣1分.

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題號 8(a)

6分

$$f(x) = x^4 - 2x^2 \text{ on } [-2, 3]$$

$$f'(x) = 4x^3 - 4x \stackrel{\text{set}}{=} 0 \quad -3 \text{分}$$

$$\Rightarrow 4x(x-1)(x+1) = 0$$

$$\Rightarrow x = -1, 0, 1 \quad -1 \text{分}$$

	-2	-1	0	1	3
f'	-	+	-	+	
f	↘	↗	↘	↗	

f is increasing on $(-1, 0) \cup (1, 3) \quad -1 \text{分}$
 decreasing on $(-2, -1) \cup (0, 1) \quad -1 \text{分}$

• 區間寫超出 domain, 一個扣一分

• 只微分, 沒有要令 $f(x) = 0$ 的動機, 得兩分.

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題號 8(b)(i)

(b) By (a), critical number: $x = 0, \pm 1$ 1pt

$$f''(x) = 12x^2 - 4 = 12\left(x - \frac{1}{\sqrt{3}}\right)\left(x + \frac{1}{\sqrt{3}}\right)$$

$\therefore f''(0) = -4 < 0 \therefore$ $f(0) = 0$ is relative max. 1pt 1pt 1pt

$\therefore f''(-1) = 8 > 0 \therefore$ $f(-1) = -1$ is relative min. 1pt 1pt 1pt

$\therefore f''(1) = 8 > 0 \therefore$ $f(1) = -1$ is relative min. 1pt 1pt 1pt

Consider endpoints

$f(-2) = 8$ is relative max.

$f(3) = 6$ is relative max.

不計入分數

(c) $f''(x) = 12x^2 - 4 = 0 \Rightarrow x = \pm \frac{1}{\sqrt{3}}$ 1pt 1pt

$\therefore f''(x) > 0$ for $x \in (-2, \frac{1}{\sqrt{3}}) \cup (\frac{1}{\sqrt{3}}, 3)$

$\therefore f$ is concave upward on $(-2, \frac{1}{\sqrt{3}}) \cup (\frac{1}{\sqrt{3}}, 3)$ 2pts

$\therefore f''(x) < 0$ for $x \in (\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$

$\therefore f$ is concave downward on $(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$ 2pts

* 沒有考慮定義域 $[-2, 3] \rightarrow$ 扣 2 分.

9. 改分依據

(a)、(b) 都有寫，只改 (a) by 應試規則 10.
如書寫內容有誤，或未交待邏輯，不給分

(a) given $\varepsilon > 0$, $\exists \delta > 0$ s.t. if $0 < |x - x_0| < \delta$
then $|f(x) - L| < \varepsilon$ ✗

(b) For $|3x+1-7| < \varepsilon \quad \forall \varepsilon > 0$, we want to find $\delta > 0$
s.t. if $0 < |x-2| < \delta$, then $|3x+1-7| < \varepsilon$

We see $|3x+1-7| = |3x-6| = 3|x-2|$.

Therefore as we choose $\delta = \frac{\varepsilon}{3}$, ($0 < |x-2| < \frac{\varepsilon}{3}$)

$|3x+1-7| = 3|x-2| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$.

Finally, given $\varepsilon > 0$, we can choose $\delta = \frac{\varepsilon}{3}$ s.t.
if $0 < |x-2| < \delta = \frac{\varepsilon}{3}$, then $|3x+1-7| < \varepsilon$ ✗